

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

***CO1:** Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Mohammed Mudhassir A / 192424367 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Mudhassir

1. Machine Learning System

Machine Learning Pipeline:

Data Collection → Data Preprocessing → Feature Engineering → Model Selection → Model Training → Model Evaluation → Deployment.

Blocks:

- Data Collection: Gather data from databases, sensors, websites.
- Data Preprocessing: Clean data, remove missing values, normalize.
- Feature Engineering: Select/create important features.
- Model Selection: Choose suitable algorithm.
- Model Training: Learn patterns from training data.
- Model Evaluation: Measure Accuracy, Precision, Recall, F1-score.
- Deployment: Use the trained model for predictions.

Supervised Learning Example: Spam email classification using labeled emails.

Unsupervised Learning Example: Customer segmentation using purchasing behavior.

2. Disaster Management

Use Random Forest or LSTM with rainfall, river level, weather and seismic data.

Preprocess by cleaning, normalization and feature selection. Train on historical data, validate on validation data and evaluate using Accuracy/Recall (or RMSE for regression).

The system predicts disasters early, sends alerts and supports evacuation and emergency response.

3. Find-S (Play Tennis)

Initial $S = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

Positive examples:

$E3 \rightarrow \langle \text{Overcast}, \text{Hot}, \text{High}, \text{Weak} \rangle$

$S = \langle \text{Overcast}, \text{Hot}, \text{High}, \text{Weak} \rangle$

$E4 \rightarrow S = \langle ?, ?, \text{High}, \text{Weak} \rangle$

$E5 \rightarrow S = \langle ?, ?, ?, \text{Weak} \rangle$

$E7 \rightarrow S = \langle ?, ?, ?, ? \rangle$

Final Hypothesis: $S = \langle ?, ?, ?, ? \rangle$

4. Candidate Elimination

Initialize $S = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$, $G = \langle ?, ?, ?, ? \rangle$.

After processing all examples:

Final $S = \langle \text{Sunny}, ?, ?, \text{Weak} \rangle$

Final $G = \{ \langle \text{Sunny}, ?, ?, ? \rangle \}$.

5. Find-S (Customer Buys Product)

Initial $S = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

$E3 \rightarrow \langle \text{Middle}, \text{High}, \text{No}, \text{Fair} \rangle$

$E4 \rightarrow S = \langle ?, ?, \text{No}, \text{Fair} \rangle$

Final $S = \langle ?, ?, \text{No}, \text{Fair} \rangle$.

6. Find-S (Workout)

Initial $S = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

$E2 \rightarrow \langle \text{Rainy}, \text{Mild}, \text{Evening}, \text{Happy} \rangle$

$E3 \rightarrow \langle ?, \text{Mild}, \text{Evening}, \text{Happy} \rangle$

$E4 \rightarrow \langle ?, ?, ?, \text{Happy} \rangle$

Final $S = \langle ?, ?, ?, \text{Happy} \rangle$.

7. Find-S (Loan Approval)

Initial $S = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$

$E1 \rightarrow \langle \text{Employed}, \text{High}, \text{Good}, \text{Clean} \rangle$

$E2 \rightarrow \langle \text{Employed}, ?, \text{Good}, \text{Clean} \rangle$

$E4 \rightarrow \langle \text{Employed}, ?, ?, \text{Clean} \rangle$

Final $S = \langle \text{Employed}, ?, ?, \text{Clean} \rangle$.

8. Candidate Elimination (Subscription)

Initialize $S = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$, $G = \langle ?, ?, ?, ? \rangle$.

Final $S = \langle \text{Young}, ?, \text{Frequent}, \text{Mobile} \rangle$

Final $G = \{ \langle \text{Young}, ?, ?, ? \rangle, \langle ?, ?, ?, \text{Mobile} \rangle \}$.

9. Candidate Elimination (Healthy Lifestyle)

Initialize $S = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle$, $G = \langle ?, ?, ?, ? \rangle$.

Final $S = \langle \text{Balanced}, ?, ?, ? \rangle$

Final $G = \{ \langle \text{Balanced}, ?, ?, ? \rangle \}$.

10. Health Analysis

Use Random Forest/Decision Tree/Logistic Regression with age, BP, glucose, BMI and lifestyle features. Preprocess by cleaning, encoding and normalization. Train-test split (80:20), evaluate using Accuracy, Precision, Recall and F1-score.

11. ML Pipeline

Pipeline: Data Collection $\rightarrow$ Preprocessing $\rightarrow$ Feature Engineering $\rightarrow$ Training $\rightarrow$ Evaluation $\rightarrow$ Deployment.

Training error is error on training data; generalization error is error on unseen test data.

Large difference indicates overfitting.

12. Spam Email Classification

Algorithm: Naive Bayes.

Preprocessing: Tokenization, lowercasing, stop-word removal, stemming/lemmatization, TF-IDF.

Metrics: Accuracy, Precision, Recall and F1-score.

13. Candidate Elimination Theory

Initialize S with most specific hypothesis and G with most general hypothesis. Positive examples generalize S, negative examples specialize G. Candidate Elimination maintains the complete version space, whereas Find-S returns only one maximally specific hypothesis.

14. House Price Prediction

Preferred model: Random Forest Regression. Handle missing values using mean/median imputation, scale features when required, detect overfitting using validation performance, and reduce it using cross-validation, pruning, regularization and early stopping.

15. Clustering

Clustering groups similar data without labels. K-Means: choose K, initialize centroids, assign points, recompute centroids and repeat until convergence. Elbow Method selects optimal K using WCSS. K-Means is fast and partition-based, whereas Hierarchical Clustering builds a dendrogram and does not require predefined K initially.